DRP write up

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Abstract

Presents some results found in [1]

1 Introduction

Theorem 1 (Absolutely continuous w.r.t Lebesgue measure)**.** *We say that a measure μ on $\mathcal{B}(\mathbb{R}^d)$ is absolutely continuous with respect to the Lebesgue measure λ if for all $A \in \mathcal{B}(\mathbb{R}^d)$ and $m(A) = 0$ then $\mu(A) = 0$.*

Corollary 2. *Let μ be absolutely continuous with respect to the Lebesgue measure, then for any $A \in \mathcal{B}(\mathbb{R}^d)$, denoting the density function of μ by f we have*

$$\mu(A) = \int_A f dx$$

Definition 3 (Random Voronoi Tesselation)**.** Let $X_1, \ldots, X_n$ be i.i.d random vectors taking values in $\mathbb{R}^d$. That means all of them share a common distribution , call it μ. We assume that μ is absolutely continuous with respect to the Lebesgue measure λ, and denote the density function of μ by f. These X_i's define a partition of $\mathbb{R}^d$ into n sets $S_1, \ldots, S_n$ such that S_i contain all points in $\mathbb{R}^d$ whose distances to X_i is not greater than their distances to all other X_j's ($j \neq i$). Letting $\|.\|$ denote the standard Euclidean norm, we have that

$$S_i = \{x \in \mathbb{R}^d \colon \|x - X_i\| = \min_{j=[n]} \|x - X_j\|\} \cap \{x \in \mathbb{R}^d \colon \|x - X_i\| < \min_{j=1,\ldots,i-1} \|x - X_j\|\}$$

Proposition 4 (Asymptotic second moment)**.** *The asymptotic second moment is expressed in terms of a random variable W defined as follows. Let Y be a random vector uniformly distributed in $B(0,1)$. Define, $\vec{1} = (1,0,0,\ldots,0) \in \mathbb{R}^d$ and let $\overline{B} = B(\vec{1},1) \cup B(Y, \|Y\|)$. Is this assuming standard Euclidean norm?*

We introduce the random variable

$$W = \frac{m(\overline{B})}{m(B(0,1))}$$

and let

$$\alpha(d) \equiv \mathbb{E}\left[\frac{2}{W^2}\right]$$

Theorem 5. *Assume μ has density f. Then*

$$n\mathbb{E}[\mu(S_1) \colon X_1 = x] \to 1$$

for μ almost all x.

Proof. Observe that

$$\mathbb{E}(\mu(S_1) \colon X_1 = x) = \mathbb{P}\{X_{n+1} \in S_1 \colon X_1 = x\}$$

Where we used a neat trick to interchange $\mu(S_1)$ with $X_{n+1} \in S_1$ since X_{n+1} is μ distributed the probability of it landing in S_1 is the same as the μ measure of S_1. Furthermore we have that

$$\mathbb{P}(X_{n+1} \in S_1 \colon X_1 = x) = \mathbb{P}\left(\bigcap_{i=2}^{n} \{X_i \notin B\left(X_{n+1}, \|X_{n+1} - x\|\right)\}\right)$$

This is because since $X_{n+1} \in S_1$ so it must be at least $\|X_{n+1} - x\|$ distance away from any other X_i $(i \geq 2)$. Defining $Z(x) = \mu(B(X, \|X - x\|))$ and since all the X_i's are indenpendently distributed, we get that

$$\mathbb{P}\left(\bigcap_{i=2}^{n} \{X_i \notin B\left(X_{n+1}, \|X_{n+1} - x\|\right)\}\right) = \mathbb{E}[(1 - Z(x))^{n-1}]$$

Now we build some intuition on the distribution of $\mu(B(x, \|X - x\|)$, Letting $Y = \mu(B(x, \|X - x\|))$ and F_Y denote the cumulative density function we know that latter is continuous and strictly increasing Why? Because of Lebesgue differentiability? which means it has a continuous injective inverse, and since F_Y^{-1} is surjective, it's bijective thus F_Y is invertible. Observing that

$$\mathbb{P}(Y \leq z) = \mathbb{P}(x \leq F_Y^{-1}(z)) = F_Y(F_Y^{-1}(z)) = z$$

So Y is uniformly distributed on $[0, 1]$

$\square$

$Z(x) = \mu(B(X, \|X - x\|))$

References

[1] Luc Devroye, László Györfi, Gábor Lugosi, and Harro Walk. On the measure of voronoi cells. 2015.